MINOR PROJECT 1

# LOAN APPROVAL PREDICTION SYSTEM

Artificial Intelligence and Machine Learning (AIML) Track

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| Department / Branch | Computer Science and Engineering (CSE Core) |
| Academic Session | 2024–2028 |
| Faculty Guide | Not provided |

GitHub: https://github.com/rahulkt2411/loan-approval-prediction-system

Live application: https://loan-approval-prediction-system-8dn6o5usdvqfd2y8tvmcjw.streamlit.app/

## Declaration

I, RAHUL KUMAR, declare that this report describes my Minor Project 1 titled “Loan Approval Prediction System” for the AIML track at RUNGTA COLLEGE OF ENGINEERING AND TECHNOLOGY. The project links and screenshots are included for review. Dataset details and quantitative model-evaluation metrics are not asserted unless they are verified from the actual notebook output.

Student Signature: _____ Date: _____

## Acknowledgement

I express my gratitude to RUNGTA COLLEGE OF ENGINEERING AND TECHNOLOGY for providing the opportunity to undertake this project. I acknowledge the open-source Python, scikit-learn, Streamlit, and joblib communities and the dataset provider used in the notebook. Faculty guide details may be added if applicable.

## Abstract

This project presents LoanWise, a web-based loan approval prediction prototype built for academic demonstration. The application allows a user to enter applicant profile and loan details, request a prediction, and view a model-estimated approval probability. The supplied application screenshots show an applicant form, a prediction result of “Likely approved” with a displayed probability of 63.3% for the example inputs, and a separate monthly EMI calculator. The application interface identifies the probability as a model estimate and states that it is not a guarantee or actual bank decision. Logistic Regression is the stated model for this project. The displayed 63.3% is one example prediction, not a model accuracy score.

## 1. Introduction

Loan applications are assessed using applicant and loan-related attributes. Supervised machine learning can learn associations between historical labelled examples and outcomes, then estimate a class for a new record. This project packages a loan-status classifier into a Streamlit application to make the prediction workflow accessible through a browser.

The application is an educational prototype. Its outputs should not be used as the sole basis for real credit, financial, or lending decisions.

## 2. Problem Statement

To develop an interactive web application that accepts applicant and loan details, uses a Logistic Regression classification model to estimate loan approval status, and presents the prediction to the user.

## 3. Objectives

• Provide a simple browser-based interface for loan-related input fields.

• Use Logistic Regression as the stated classification approach.

• Load a persisted model artifact for prediction in the Streamlit application.

• Display the predicted outcome and, where available, a model-estimated probability.

• Provide an EMI calculator as a separate financial calculation utility.

• Document the project, verify application behavior, and report only measured evaluation results.

## 4. Scope and Technology Stack

The project scope covers model-based classification and an interactive demonstration interface. The supplied screenshots show navigation for Dashboard, Make a Prediction, EMI Calculator, Model Performance, Feature Importance, Prediction History, and About Project. Only behavior actually implemented in the current deployed version should be claimed in a final viva or report.

• Programming language: Python (version to confirm from project environment).

• Machine-learning model: Logistic Regression (as provided by the student).

• Web application: Streamlit.

• Model persistence: joblib model artifact (repository file: loan_approval_model.joblib).

• Development environment: Google Colab.

• Version control and source hosting: GitHub.

## 5. Dataset and Input Features

Dataset name, provider, licence, size, exact target column, feature list, missing-value treatment, categorical encoding, feature scaling, and train/test strategy must be copied from the actual Colab notebook. These details are intentionally not guessed in this report.

| Item | Details to verify from notebook |
|---|---|
| Dataset title and source | [Enter exact title, URL and licence] |
| Records and columns | [Enter actual counts] |
| Target column / labels | [Enter exact target and class mapping] |
| Model input features | [Enter actual ordered feature names] |
| Preprocessing | [Describe only implemented steps] |
| Evaluation method | [Train/test split or cross-validation details] |

## 6. Methodology

• Data inspection: review the dataset columns, target labels, missing values, and data quality.

• Data preparation: apply the preprocessing actually implemented in the notebook.

• Model training: train Logistic Regression on the prepared training data, if training occurs in the notebook.

• Evaluation: use the notebook's documented holdout or validation approach and record measured metrics.

• Model persistence: save and load the trained estimator or pipeline using joblib.

• Application inference: collect form values, construct input data in the expected feature schema, and request a prediction.

• Presentation: display the prediction and model-estimated probability when supported by the model.

• Additional utility: calculate a hypothetical loan EMI separately from the ML prediction.

## 7. System Architecture

**Data source → preprocessing → Logistic Regression training → evaluation → saved model artifact → Streamlit form → prediction and probability display.**

The EMI calculator is a separate component: loan principal, annual interest rate, and tenure are used to calculate an estimated monthly instalment, total payment, and total interest. This calculation is not a model prediction.

## 8. Application Screenshots

The following screenshots were supplied by the student from the deployed Streamlit application.

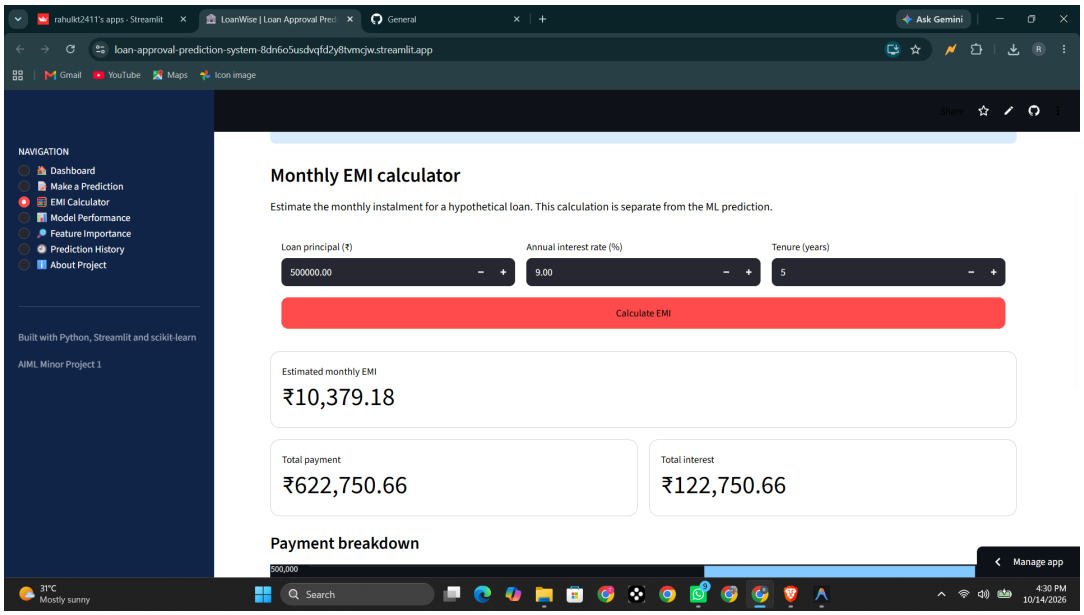



Figure 1. Applicant profile and loan details input form.

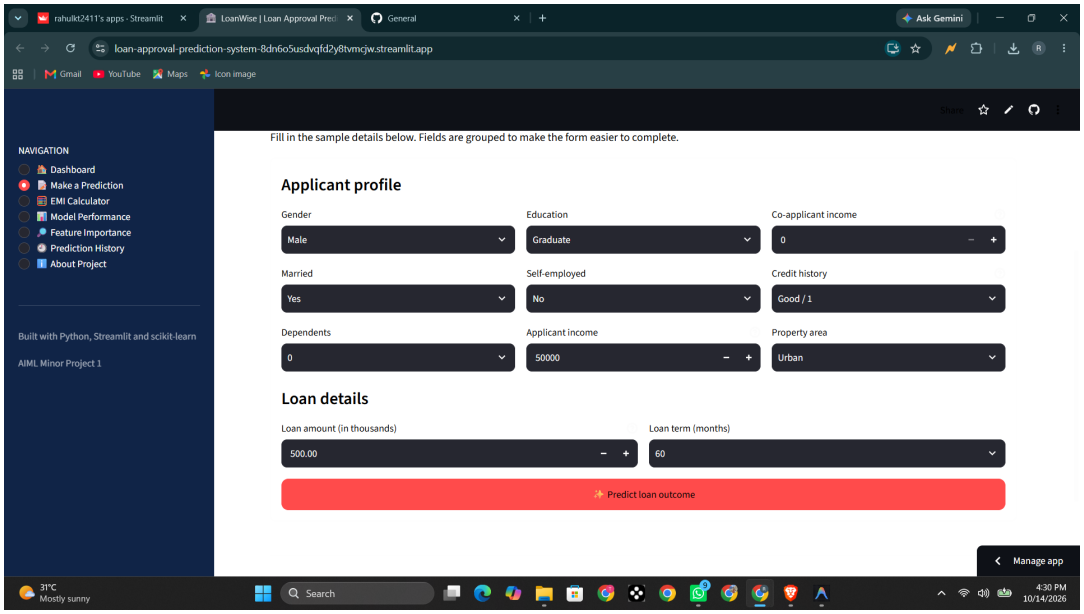



Figure 2. Example prediction result: “Likely approved” and 63.3% model-estimated approval probability.

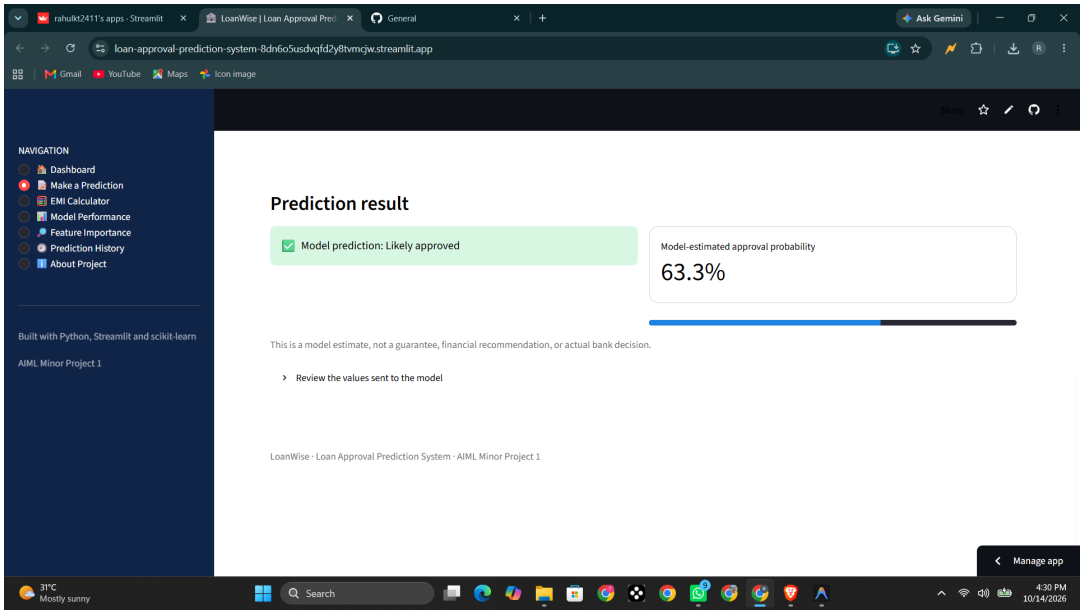



Figure 3. EMI calculator example showing principal ₹5,00,000, annual interest 9%, tenure 5 years, and estimated EMI ₹10,379.18.

## 9. Testing

| Test case | Evidence / expected outcome | Status |
|---|---|---|
| Prediction form | Form fields visible in supplied screenshot | Observed in screenshot |
| Example prediction | Result and 63.3% probability displayed | Observed in screenshot |
| EMI calculator | EMI, total payment and interest displayed | Observed in screenshot |
| Model Performance / Feature Importance | Navigation entries visible; detailed behavior not verified | Verify live app |
| Model artifact loading | Repository includes a saved model artifact per prior repository review | Verify by running app |
| Invalid input handling | Not evidenced by supplied screenshots | Test before submission |

## 10. Model Evaluation and Results

The supplied screenshot verifies one inference example, not overall model performance. A displayed approval probability of 63.3% must not be reported as 63.3% accuracy. The evaluation-results CSV was previously observed with metric headings but no populated result rows in its public preview. Therefore, this report does not state accuracy, precision, recall, F1-score, ROC-AUC, or a confusion matrix.

Before final submission, run the evaluation cells in the Colab notebook and record only metrics actually calculated on the stated test/validation data. Explain which label is considered the positive class and describe the split or validation method.

## 11. Limitations and Responsible Use

• Predictions depend on the training dataset, feature definitions, preprocessing, and representativeness of the examples.

• Historical lending data may contain bias; model performance should be assessed beyond overall accuracy.

• A predicted probability is a model output and may not be calibrated to real-world approval rates.

• Input preprocessing in the deployed app must match training-time preprocessing and feature ordering.

• The EMI calculator provides a mathematical estimate and may exclude fees, insurance, taxes, or lender-specific terms.

• This application is an academic prototype, not a real credit approval or financial advice system.

## 12. Conclusion and Future Scope

LoanWise demonstrates an end-to-end concept for presenting a loan approval classification model through a Streamlit interface. The supplied screenshots show the input form, an example prediction with a model-estimated probability, and a separate EMI calculator. Quantitative claims about predictive performance remain pending a verified evaluation run.

• Populate a reproducible evaluation report and confusion matrix.

• Validate inputs and add clear error messages for unsupported values.

• Review probability calibration and fairness limitations before any real-world interpretation.

• Add dataset attribution, versioned model training steps, and deployment instructions.

## 13. References


• Python documentation: https://docs.python.org/3/

• scikit-learn documentation: https://scikit-learn.org/stable/

• Streamlit documentation: https://docs.streamlit.io/

• GitHub repository: https://github.com/rahulkt2411/loan-approval-prediction-system

• Google Colab notebook: https://colab.research.google.com/drive/1HWvlstT-uvcoJwy5BrhV5cRekh0WQ8lC#scrollTo=P9UzhP-ZUfxz

• Dataset citation: [add the actual dataset provider, URL, licence, and access date].


## Appendix A. Project Links

GitHub repository: https://github.com/rahulkt2411/loan-approval-prediction-system

Google Colab notebook:
https://colab.research.google.com/drive/1HWvlstT-uvcoJwy5BrhV5cRekh0WQ8lC#scrollTo=P9UzhP-ZUfxz

Live Streamlit application: https://loan-approval-prediction-system-8dn6o5usdvqfd2y8tvmcjw.streamlit.app/

Submission reminder: fill the dataset details and actual metrics, check the department/college's required format, then upload this report and PPT to the accessible repository or Drive folder.